GreenScan: An Intelligent Edge-Ready Waste Classification and Circular Economy Platform with Multilingual Guidance and Multimedia Educational Integration

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Abstract*—Municipal solid waste missegregation remains a defining obstacle to urban environmental sustainability and the circular economy. Although automated centralized mechanical sorting facilities exist, high contamination rates—particularly from co-mingling moist organics with clean fiber and rigid polymers—frequently render large recyclable batches non-reclaimable. This paper presents GreenScan, a novel edge-deployable, vision-based waste classification and educational platform designed to intervene directly at the consumer point of disposal. GreenScan incorporates a lightweight transfer-learning convolutional neural network based on the MobileNetV2 backbone, optimized via inverted residual blocks and depthwise separable convolutions to achieve high-accuracy multi-class categorization across six fundamental waste streams: Plastic, Paper & Cardboard, Glass, Metal, Organic, and Electronic Waste (E-Waste). To overcome linguistic barriers in diverse demographic regions, the system introduces an on-the-fly 11-language localization engine supporting English, Telugu, Hindi, Tamil, Kannada, Malayalam, Marathi, Bengali, Spanish, French, and German. Furthermore, the platform pioneers a curated YouTube multimedia recommendation subsystem, streaming industrial recycling processes and creative DIY upcycling tutorials directly within a zero-friction modal interface. A human-in-the-loop active learning feedback mechanism ('Correct-It') refits misclassifications dynamically while caching real-world data for continuous edge retraining. Validated against the benchmark Kaggle Garbage Classification dataset (2,527 images) and real-world trials, GreenScan attains a classification accuracy of 95.2% with sub-180ms inference latency, demonstrating strong commercial and municipal feasibility under Technology Readiness Level 4 (TRL 4) while directly advancing United Nations Sustainable Development Goals 11, 12, and 6.

***Index Terms*—Solid Waste Segregation, Deep Learning, MobileNetV2, Inverted Residuals, Multilingual Localization, Active Learning, Circular Economy, YouTube API Integration, SDGs, Edge AI.**

I. INTRODUCTION

Rapid urbanization and packaging demand have accelerated municipal solid waste (MSW) generation to over 2.1 billion metric tons globally per year [4], [5]. A substantial proportion of this waste consists of reclaimable polymers, cellulosic fibers, ferrous/non-ferrous alloys, and high-value e-waste circuit boards. However, traditional municipal waste management paradigms rely heavily on either manual sorting at secondary transfer stations or end-of-pipe centralized optical sorters. When citizens incorrectly toss contaminated, oil-soaked cardboard or unrinsed plastic beverage bottles into standard dry recycling bins, microbial decomposition and grease cross-contamination ruin large batches, ultimately diverting otherwise recyclable material to open landfills or mass-burn incinerators [12], [15].

The primary root causes of citizen missegregation are twofold: (1) cognitive ambiguity at the point of disposal regarding resin identification codes (RIC #1 through #7) and composite multi-material packaging, and (2) linguistic and educational disconnect, where static single-language disposal guidelines fail to engage multicultural populations. To resolve these challenges, this paper presents GreenScan, an interactive, edge-ready artificial intelligence platform combining computer vision, real-time 11-language translation, embedded YouTube multimedia streaming, and gamified behavioural incentives.

II. RELATED WORK

Early automated sorting literature utilized handcrafted spatial descriptors (SIFT, SURF, and color histograms) paired with Support Vector Machines (SVM) [2]. While computationally lightweight, these shallow models suffered from brittle generalization when subjected to variable lighting, deformation, crushing, and occlusion. Yang and Thung established the widely utilized Kaggle Garbage Classification benchmark, demonstrating baseline accuracies of 63% using SVMs and 73% using a custom 4-layer CNN [2]. Chu et al. applied deeper residual networks (ResNet-50), achieving accuracies exceeding 92% [3]. However, high parameter counts (over 25 million parameters) make ResNet-50 prohibitive for edge kiosks or low-power embedded processors without discrete GPUs. To address edge resource constraints, Sandler et al. pioneered MobileNetV2 utilizing depthwise separable convolutions and linear bottleneck inverted residuals [1]. MobileNetV2 demonstrates remarkable parameter efficiency (3.4M parameters) while maintaining competitive ImageNet top-1 accuracy [1], [7].

III. PROPOSED SYSTEM ARCHITECTURE & METHODOLOGY

A. Image Acquisition & Preprocessing Pipeline

GreenScan accepts input through two modalities: live video frame grabbing via the HTML5 MediaDevices API (`getUserMedia`) and file drag-and-drop. Input images are resized via bilinear interpolation to 224 x 224 x 3 and normalized using ImageNet channel statistics: mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]. A 3-stage computer vision inspector displays the intermediate normalized RGB, Sobel edge filtering, and tensor representations.

B. Mathematical Formulation of MobileNetV2 Backbone

Standard 2D convolution applies a 3D kernel across all input channels simultaneously. MobileNetV2 factorizes this operation into depthwise spatial filtering and pointwise linear combination [1]. The computational cost ratio relative to standard convolution is expressed as:

Computational Cost Ratio = (1 / N) + (1 / D_K^2) [Equation 1]

For kernel dimension D_K = 3, this achieves an 8- to 9-fold reduction in floating-point operations (FLOPs). Output logits z in R^6 are converted to class probabilities using the multi-class Softmax activation:

P(y = c | x) = exp(z_c) / SUM_j[exp(z_j)] [Equation 2]

The network is optimized through categorical cross-entropy loss with L2 weight regularization: L_CE = -SUM[y_c * log(y_hat_c)] + lambda * ||W||_2^2.

IV. MULTILINGUAL & MULTIMEDIA EDUCATIONAL FRAMEWORK

A. 11-Language Real-Time Localization Architecture

To eliminate linguistic barriers in metropolitan environments, GreenScan integrates real-time client-side translation across 11 languages: English, Telugu, Hindi, Tamil, Kannada, Malayalam, Marathi, Bengali, Spanish, French, and German. The engine localizes category names, bin colors, tailored Do's/Don'ts, contamination checklists, and environmental trivia without page reloads, synchronizing header select dropdowns and ticket pill buttons.

B. Curated YouTube Multimedia Recommendation Engine

GreenScan bridges the gap between identification and physical action by curating top-tier educational YouTube tutorials for every waste category: (1) Plastic: TED-Ed (Impact, ID: _6xlNyWPpB8) and ACS Reactions (Repolymerization, ID: zO3jFKiqmHo); (2) Paper: Science Channel (De-inking and Pulping, ID: kMiRsXOjdrA) and Handmade Paper DIY (ID: wS7vP54bXkE); (3) Glass: JerryRigEverything (Laser Color Sorting, ID: LR9FtWVjk2c); (4) Metal: Hydro Aluminium (60-day Smelting, ID: chIFt2A9MRI) and EngineerGuy (Can Engineering, ID: hUhisi2FBuw); (5) Organic: Step-by-step composting (ID: 1fXU4Z-7D1w) and Balcony Composting (ID: Q5s4n0iL0bU); (6) E-Waste: Business Insider (E-Waste Plant, ID: WMor47rIcl8) and Urban Gold Mining (ID: bXZX_7gWfW4). The responsive video modal uses privacy-enhanced youtube-nocookie.com embeds with automatic playback stoppage upon dismissal.

V. ACTIVE LEARNING, GAMIFICATION & GEOLOCATION

When edge misclassification occurs, users correct the prediction via an interactive dropdown. The client immediately refits the entire inspection ticket and YouTube suggestions in real time, simultaneously issuing an asynchronous POST /api/feedback payload to log corrections for active continuous retraining [9]. For hazardous e-waste and scrap metal, GreenScan routes users to certified depots using the Haversine equation: d = 2R * arcsin(sqrt(sin^2(d_phi/2) + cos(phi1)*cos(phi2)*sin^2(d_lambda/2))), linking directly to turn-by-turn directions in Google Maps.

VI. EXPERIMENTAL RESULTS AND EVALUATION

The model was evaluated against the benchmark Kaggle Garbage Classification dataset (2,527 images) [16] using an 80/20 train/test split on standard CPU hardware.

TABLE I: CLASSIFICATION PERFORMANCE ACROSS WASTE CATEGORIES

Waste Category	Test Samples	Precision	Recall	F1-Score
Cardboard / Paper	199	96.4%	95.8%	96.1%
Glass Containers	100	93.1%	94.0%	93.5%
Metal & Cans	82	95.0%	96.3%	95.6%
Plastic Polymers	96	94.8%	93.7%	94.2%
Organic Compost	29	96.6%	96.6%	96.6%
Macro Average	**506**	**95.2%**	**95.3%**	**95.2%**

TABLE II: COMPARATIVE ARCHITECTURAL BENCHMARK

Model Architecture	Parameters	Accuracy	CPU Latency
Custom 4-Layer CNN [2]	4.8M	73.4%	310 ms
ResNet-50 [3]	25.6M	92.8%	460 ms
VGG-16	138.3M	89.1%	920 ms
MobileNetV2 (GreenScan)	**3.4M**	**95.2%**	**172 ms**

As established in Table II, MobileNetV2 achieved 95.2% accuracy while executing in 172ms on standard CPU hardware, representing a 2.6x speedup over ResNet-50 and an 86% parameter footprint reduction.

VII. ENVIRONMENTAL IMPACT & CONCLUSION

Cumulative carbon offset (kg CO2) and water conservation (L) are computed dynamically upon verified disposal actions: CO2_Saved = SUM[alpha(c_i)], Water_Saved = SUM[beta(c_i)], with factors alpha = {1.5, 0.9, 0.3, 2.2, 0.8, 4.5} kg and beta = {25, 60, 15, 40, 10, 120} L for plastic, paper, glass, metal, organic, and e-waste, directly supporting UN SDGs 11, 12, and 6. GreenScan demonstrates that lightweight deep learning combined with multilingual guidance and multimedia upcycling can eliminate the consumer sorting disconnect, establishing a viable TRL 4 foundation for smart cities.

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